Liana Valdés

Doctorate in Computer Science.
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Education

Florida International University — **21 August 2017 - 16 December 2023**
Doctor of Philosophy in Computer Science — GPA: 3.83/4
Advisor: *Eminent Scholar Chaired Professor Raju Rangaswami.*
Relevant courses: *Analysis of Algorithms, Introduction to Algorithms, Advanced Software Engineering, Database Management Systems, Introduction to Machine Learning, Data Visualization, Operating Systems, Theory of Computation, Computer Communication and Networking Technologies, Secure Application Programming.*

Florida International University — **21 August 2017 - 17 December 2022**
Master of Science in Computer Science — GPA: 3.83/4

Technological University of Havana "José Antonio Echeverría" — **1 September 2009 - 20 July 2014**
Bachelor of Science in Telecommunications and Electronics Engineering — Havana, Cuba
Faculty of Telecommunications Engineering — GPA: 4.48/5

Experience

Graduate Research Assistant/Teaching Assistant/GAANN Fellowship | Florida International University, FL, USA August 2017 - December 2023
- *Developed LeCaR and CACHEUS (94.38% hit rate), ML systems for cache replacement, CaaS, a novel distributed caching, and TxFuse.*

Research Intern | Seagate Technology, Remote, USA — **August 2021 - December 2021**
- *Tested the CORTX's software stack, the File Data Manipulation Interface (FDMI), and conducted a performance analysis on Motr storage.*

Research Intern | Microsoft Research (MSR), Cambridge, England, UK — **January 2020 - March 2020**
- *Developed ML systems using DL and Unsupervised Learning to identify anomalies when tested using image data from the movie "Superman".*

Telematics Specialist B | Cuban Telecommunications Company S.A., Havana, Cuba — **September 2015 - June 2016**
- *Maintained uninterrupted communication services on IT infrastructures.*

Research Intern/Apprenticeship | Cuban Radio and Television Broadcasting Company, Havana, Cuba — **March 2012 - July 2014**
- *Analyzed the board layout to modify the UHF and VHF communication modules from the PAL television standard to NTSC.*

Research Projects

Extending storage systems to meet cloud needs | Seagate Technology & SyLab (FIU) — **August 2021 - Present**
- *Designed TxFuse, a novel architecture, to extend a distributed storage system with plugins and improve performance and efficiency.*
- *Developed transactional coupling and reliable notification techniques to implement storage features such as encryption and compression.*

A distributed cache for cloud data centers | SyLab, ModLab, DAMRL (FIU) & HASLab (UMinho) — **May 2020 - Present**
- *Designed and implemented CaaS, a novel, distributed, and generalized cache for cloud computing infrastructures using novel QoS algorithms.*
- *Improved hit rates, data center services, and microservices with a distributed cache that unifies cache resources for reading and writing.*

ML systems for caches in the cloud | SyLab, ModLab, BioRG (FIU) & VISA (ASU) — **August 2018 - February 2021**
- *Developed ML algorithms LeCaR and CACHEUS to improve cache hit rates compared to classic cache replacement algorithms for the cloud.*

Honors & Awards

USENIX Student Travel Award, FAST'18 & FAST'19, & FAST'23.
GAANN Fellowship from the ED, 2022 and 2023.
CMD-IT/ACM Richard Tapia Celebration of Diversity, 2022 and 2024.
Grace Hopper Celebration of Women in Computing, 2019 and 2022.
USENIX SREcon24 Europe/Middle East/Africa Award, 2024.
Reviewer nationwide for ACM Transactions on Storage (TOS)'21,'25.
***Google Scholar Profile**: Link*

Skills

Cache,Storage,ML,RL,DL,SL UL,AI, JavaScript,TypeScript,Keras Distributed Systems,Python,C#,C,C++,Java,Go,R,R++,VS Code TeX,VB,HLA,TensorFlow,PyTorch,Scikit-learn,Docker,GitLab, Team Leadership,Distributed caching,QoS,CompTIA A+,Agile,Git GitHub,Tableau,PowerBI,Prometheus,Grafana,Project to develop LLM-based AI Mobile agents in AndroidWorld with optimized Prompt engineering using Gemini (GitHub links: AI agent, Prompts). Other sample ML projects (Gold-priceANN,ANN,Classifier).

Publications

Project Silica: Towards Sustainable Cloud Archival Storage in Glass — **October 2023**
Liana Valdes, in Acknowledgement, The 29th ACM Symposium on Operating Systems Principles, SOSP'23.

Infusing Pub-Sub Storage with Transactions — **July 2022**
The 14th ACM Workshop on Hot Topics in Storage and File Systems, HotStorage'22.

Unifying the Data Center Caching Layer - Feasible? Profitable? — **July 2021**
The 13th ACM Workshop on Hot Topics in Storage and File Systems, HotStorage'21.

Learning Cache Replacement with CACHEUS — **February 2021**
19th USENIX Conference on File and Storage Technologies, FAST'21.

Driving Cache Replacement with ML-Based LeCaR — **July 2018**
10th USENIX Workshop on Hot Topics in Storage and File Systems, HotStorage'18.